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SEMINAR

Women Safety Mobile Application (Raksha App)

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1. Introduction

Modern mobile applications, especially those focused on safety like the **Raksha App**, must support real-time tracking, instant alerts, secure data handling, and continuous updates. These applications are critical as they directly impact user safety and must function reliably under all conditions.

Traditional software development models such as the Waterfall model are not suitable for such dynamic systems, as they lack flexibility and slow down updates. To address these challenges, large organizations like Infosys adopt **Agile process models integrated with DevOps practices**.

Agile enables iterative development, fast feedback, and continuous improvement, while DevOps ensures automation, continuous integration, deployment, and monitoring. This report explains how Agile and DevOps are applied using the Raksha App case study and highlights the use of tools like Jira for project management.

2. Overview of Agile Process Model

The Agile process model is an iterative and incremental approach where development is divided into small cycles called **sprints**. Each sprint includes planning, development, testing, and review.

Instead of delivering the entire system at once, Agile focuses on delivering small functional parts of the application regularly.

Key Principles of Agile:

- Customer collaboration over contract negotiation
- Working software over heavy documentation
- Responding to change over following a fixed plan
- Continuous improvement through feedback

In safety apps like Raksha, Agile helps teams quickly add features such as emergency alerts and GPS tracking based on user needs.

3. Overview of DevOps

DevOps is a practice that integrates development and operations teams to automate and streamline software delivery.

It focuses on continuous processes to ensure fast and reliable deployment.

Main DevOps Activities:

- Continuous Integration (CI)
- Continuous Testing

- Continuous Deployment (CD)
- Infrastructure Automation
- Monitoring and Feedback

Using DevOps, applications like Raksha can release updates instantly and ensure the system is always operational.

4. Agile + DevOps Integration

When Agile and DevOps are combined, they create a powerful and efficient development lifecycle.

- Agile manages requirements, sprint planning, and development cycles
- DevOps automates build, testing, deployment, and monitoring

This integration ensures:

- Faster development
- Reliable deployment
- Continuous delivery of features

Large IT companies like Infosys use this combination to maintain high-quality software and meet changing user demands.

5. Case Study: Women Safety Mobile Application (Raksha App)

The **Raksha App** is a mobile application designed to ensure the safety of women by providing emergency assistance features such as instant alerts, GPS tracking, and communication with trusted contacts.

It is widely used in urban areas where quick response and real-time tracking are essential.

5.1 System Requirements of Raksha App

Functional Requirements:

- User registration and login
- SOS emergency alert button
- Real-time GPS location tracking
- Send alerts to emergency contacts
- Integration with police helpline
- Audio/video recording during emergencies
- Push notifications and alerts

Non-Functional Requirements:

- High availability (24/7 service)
- Fast response time during emergencies
- Secure user data handling
- Scalability for large user base
- Reliability and accuracy in tracking
- Data privacy and security

5.2 Agile Implementation Using Jira

Development teams use tools like Jira to manage Agile workflows.

Agile Workflow:

1. **Product Backlog** – List of features (SOS button, GPS tracking, alerts)
2. **Sprint Planning** – Selecting tasks for development
3. **Development** – Building modules like emergency alert system
4. **Testing** – Unit testing, integration testing, UI testing
5. **Sprint Review** – Demonstrating features to stakeholders
6. **Sprint Retrospective** – Improving team performance

Jira helps track tasks, assign responsibilities, and monitor sprint progress efficiently.

5.3 DevOps Implementation**DevOps Tools:**

- Project Management – Jira
- Version Control – Git
- Build Automation – Jenkins
- Containerization – Docker
- Cloud Infrastructure – AWS
- Monitoring – Prometheus / Grafana

DevOps Workflow:

1. Developers write and push code to Git
2. Jenkins builds the application automatically

3. Automated tests are executed
4. Docker packages the application
5. Deployment to AWS cloud servers
6. Monitoring tools track performance and errors

This ensures that the Raksha App is always updated and functioning properly.

6. Benefits of Agile + DevOps for Raksha App

- Faster development of safety features
- Continuous updates based on user feedback
- High reliability through automated testing
- Scalable infrastructure for increasing users
- Improved safety and user experience

7. Why Agile + DevOps is Recommended in Companies like Infosys

Companies like Infosys recommend Agile with DevOps because:

- Faster development cycles
- Better collaboration between teams
- High-quality software delivery
- Quick bug detection and fixing
- Continuous deployment and monitoring

For critical applications like Raksha, these benefits are essential to ensure safety, reliability, and performance.

8. Conclusion

Agile process models combined with DevOps practices provide an effective framework for developing modern safety applications.

The case study of the Raksha App shows how Agile and DevOps help in building scalable, reliable, and continuously improving systems. Tools like Jira further enhance project management and workflow tracking.

Therefore, Agile with DevOps is highly recommended for large organizations like Infosys that aim to develop secure, scalable, and user-focused applications.